Project Title

Résumé–Job Matching & Talent-Marketplace Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **Résumé–Job Matching & Talent Marketplace Engine** is an intelligent recruitment platform designed to simplify and improve the hiring process by automatically matching job seekers with suitable employment opportunities. Traditional recruitment methods often require recruiters to manually review hundreds of résumés, making the process time-consuming, error-prone, and inefficient. This project addresses these challenges by utilizing advanced data structures and algorithmic techniques to analyze candidate résumés, compare them with job descriptions, and recommend the most suitable candidates based on their skills, qualifications, experience, and interests.The system performs efficient text analysis by extracting important keywords from résumés and job postings using advanced string-matching algorithms. Dynamic Programming techniques are employed to measure the similarity between candidate profiles and job requirements, enabling accurate matching even when different words or phrases express similar skills. The platform also applies graph-based matching concepts to assign candidates to appropriate job opportunities while maximizing overall compatibility and minimizing mismatches. When multiple candidates compete for limited job openings, optimization strategies are used to generate fair and effective recommendations.In addition to job matching, the system provides personalized job recommendations for candidates and ranked candidate suggestions for recruiters. It reduces manual effort, speeds up the recruitment process, improves matching accuracy, and enhances the overall hiring experience for both employers and job seekers. The project demonstrates how advanced algorithmic paradigms such as **String Algorithms, Dynamic Programming, Network Flow, and Optimization Techniques** can be integrated to solve real-world recruitment challenges efficiently and at scale.This project aligns with the objectives of the **Data Structures and Algorithms–3** course by showcasing the practical application of advanced algorithms in large-scale text processing, optimization, and intelligent decision-making systems. It highlights how theoretical concepts can be transformed into a real-world solution that improves recruitment efficiency, supports data-driven hiring decisions, and creates a smart talent marketplace for the modern workforce.

Signature of the Guide